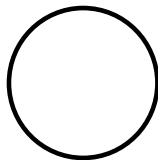




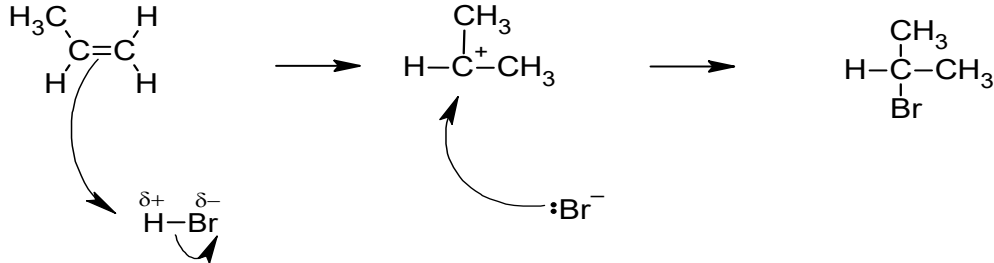
Question	Answer	Marks															
1(a)(i)	(both) giant molecular	1															
1(a)(ii)	delocalised electrons move through the structure OR delocalised electrons move through / between the layers	1															
1(a)(iii)	no delocalised electrons (to carry charge through the structure) OR <u>all</u> (4) outer electrons are used in bonding	1															
1(b)(i)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1															
	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1															
1(b)(ii)	M1 phosphorus(V) chloride white / yellow solid M2 silicon(IV) chloride colourless liquid	2															
1(b)(iii)	M1 (phosphorus(V) chloride) produces colourless solution M2 silicon(IV) chloride produces white suspension / white precipitate (in a colorless liquid / mixture / solution)	2															
1(c)(i)	species behaves as acid and base OR species reacts with (both) acid and base	1															
1(c)(ii)	Al_2O_3	1															
1(d)(i)	<table border="1"> <tbody> <tr> <td>SO_2</td><td>-73</td><td>pd</td></tr> <tr> <td>H_2O</td><td>0</td><td>H</td></tr> <tr> <td>SiO_2</td><td>1600</td><td>C</td></tr> <tr> <td>Na_2O</td><td>1132</td><td>I</td></tr> <tr> <td>MgO</td><td>2852</td><td>I</td></tr> </tbody> </table>	SO_2	-73	pd	H_2O	0	H	SiO_2	1600	C	Na_2O	1132	I	MgO	2852	I	3
SO_2	-73	pd															
H_2O	0	H															
SiO_2	1600	C															
Na_2O	1132	I															
MgO	2852	I															

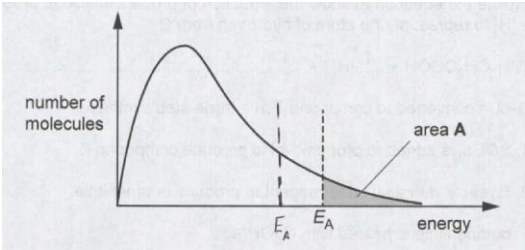
Question	Answer	Marks
1(d)(ii)	<i>Evaluate hypotheses based on the information candidates have given in the table</i> M1 not enough information M2 SO₂ and / or H₂O break intermolecular forces OR SO₂ and / or H₂O does not break covalent bonds OR only SiO₂ breaks covalent bonds	2

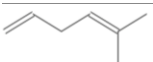
Question	Answer	Marks
2(a)(i)	route 1 M1 average (weighted) mass of (an) atom(s) (of an element) OR average (weighted) mass of the isotope(s) (of an element) M2 compared to the mass of the unified atomic mass unit OR on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 (units) OR on a scale in which the mass of a carbon-12 atom/isotope is exactly 12 (units) OR compared to $\frac{1}{12}$ mass of one atom of C-12 route 2 M1 average mass of one mol of isotopes / atoms (of an element) M2 relative / compared to $\frac{1}{12}$ of the mass of one mol of C-12 atoms/isotopes OR when the mass of one mol C-12 atoms / isotopes is exactly 12 (000) g	2
2(a)(ii)	M1 relative abundance of unknown isotope = $(100 - 91.9 - 6.0) = 2.1\%$ M2 $55.8 = (53.9 \times 6.0 + 55.9 \times 91.9 + ? \times M1) / 100 = 56.9$	2
2(b)	1 (pair)	1

Question	Answer	Marks						
2(c)		1						
2(d)	<table><tr><td>species present in an ion of 56-Fe^{3+}</td><td>number</td></tr><tr><td>electrons</td><td>M1 23</td></tr><tr><td>M2 neutrons</td><td>30</td></tr></table>	species present in an ion of 56-Fe^{3+}	number	electrons	M1 23	M2 neutrons	30	2
species present in an ion of 56-Fe^{3+}	number							
electrons	M1 23							
M2 neutrons	30							
2(e)	M1 smaller / decreases M2 loss of outer / valence shell (of electrons) OR loss of electrons so increase in (force of) attraction from nucleus OR loss of electrons so increase nuclear attraction	2						

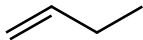
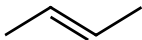
Question	Answer	Marks
3(a)(i)	two (or more) reactants / molecules react to make (only) one product	1

Question	Answer	Marks
3(a)(ii)	 M1 curly arrow from = of C=C to H (of HBr) M2 curly arrow from bond of H^{δ+}—Br^{δ-} to Br^{δ-} (dipole is essential) M3 correct intermediate M4 curly arrow from lone pair of Br⁻ to C⁺ of intermediate drawn	4
3(a)(iii)	M1 (2-bromopropane is made from the) more stable intermediate / carbocation M2 (due to positive) inductive effect of more alkyl groups (attached to the C⁺ / central carbon atom) OR (due to) greater (positive) inductive effect of 2 methyl groups (rather than 1 ethyl group)	2
3(b)(i)	H ₂ / hydrogen AND some X / alkene left(over) / remains (at the end of this reaction)	1
3(b)(ii)	<i>link the significance of gradient values to rate AND describe either the decrease in rate or the decrease in gradient</i> as the gradient decreases the rate decreases OR rate decreases until the reaction stops when gradient = 0	1

Question	Answer	Marks
3(c)	method 1 – breaking all bonds in molecules $(2 \times 610) + (2 \times 436) + (10 \times 410) + (3 \times 350) - (14 \times 410) - (5 \times 350)$ - energy involved when all bonds are broken = (7242) - energy involved when all bonds are made = (7490) - energy to break bonds – energy to make bonds $= (7242 - 7490) = -248$ method 2 – ignore any bonds that do not change from reactants to products $(2 \times 610) + (2 \times 436) - (4 \times 410) - (2 \times 350)$ - energy involved when all bonds are broken = (2092) - energy involved when all bonds are made = (2340) - energy to break bonds – energy to make bonds $(2092 - 2340) = -248$	2
3(d)(i)	E_A shown on Fig. 3.3 to the left of original (on the x axis) 	1

Question	Answer	Marks
3(d)(ii)	M1 it / (area) B larger AND rate (reaction / hydrogenation) increases OR (area) B identified by shading or labelling on Fig 3.3 AND rate (reaction / hydrogenation) increases in prose M2 increase in the frequency of effective / successful collisions OR increase in rate of effective / successful collisions	2
3(e)	$\text{H}_2\text{C}=\text{CHCH}_2\text{CH}=\text{C}(\text{CH}_3)_2$ ✓✓  ✓✓ $\text{H}_2\text{CCHCH}_2\text{CHC}(\text{CH}_3)_2$ ✓✓ M1 $\text{H}_2\text{C}=\text{C}\dots$ on one end of their molecule OR $\dots\text{C}=\text{C}(\text{CH}_3)_2$ on one end of their molecule OR $\dots\text{C}=\text{CHCH}_2\text{CH}=\text{C}\dots$ in the middle of their molecule M2 correct structure	2

Question	Answer	Marks
4(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$	1
4(a)(ii)	(ionisation energy decreases because of) • more shells / larger (atomic/ionic) radius • increased shielding • less attraction (for the nucleus/protons) / lower energy required (to remove electrons)	2
4(a)(iii)	$Ba^{2+}(g) \rightarrow Ba^{3+}(g) + e^{-}$	1
4(a)(iv)	M1 electron removed from closer / inner / lower-energy (p sub)shell/orbital M2 (much) less shielding OR greater nuclear attraction	2
4(b)(i)	M1 $Sr(s) + 2H_2O(l) \rightarrow Sr(OH)_2(aq) + H_2(g)$ M2 state symbols for <u>all</u> species	2
4(b)(ii)	M1 with HCl no visible reaction / no change (observed) M2 with H_2SO_4 white precipitate/solid forms	2
4(c)(i)	(thermal stability) increases (down the group)	1
4(c)(ii)	$2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$	1
4(d)(i)	$2OH^{-} + Cl_2 \rightarrow ClO^{-} + Cl^{-} + H_2O$	1
4(d)(ii)	(ClO^{-}) kills bacteria	1

Question	Answer	Marks
5(a)	(organic molecule where) carbon (of C—Hal) is bonded to two other C atoms	1
5(b)(i)		1
5(b)(ii)	M1  M2 <i>trans</i>-but-2-ene	2